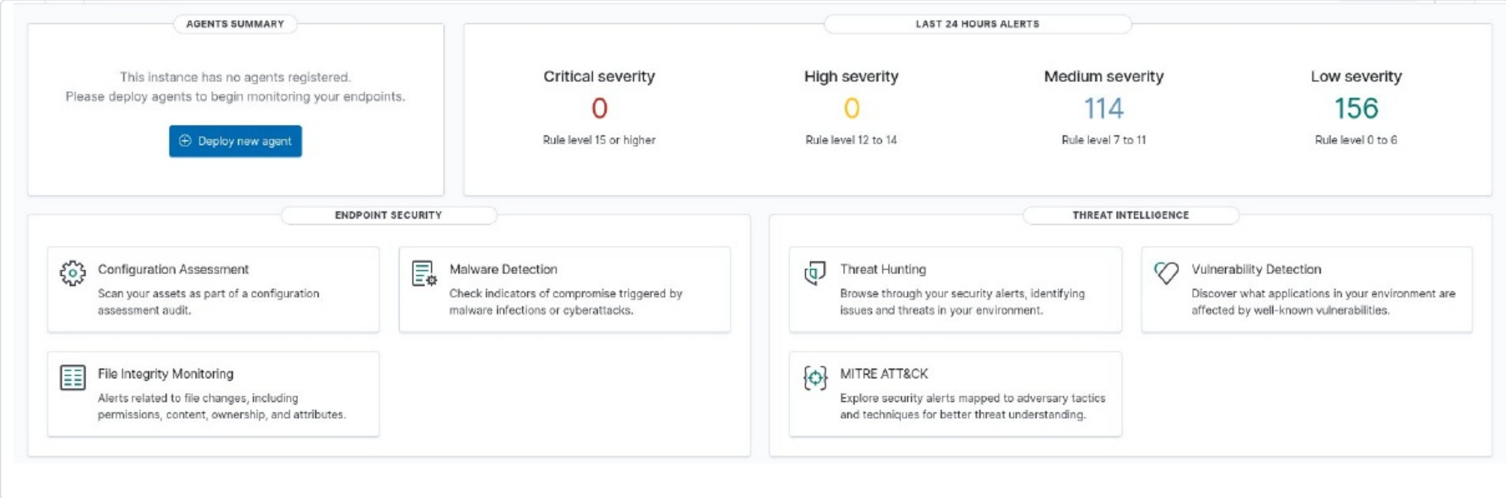
# Wazuh SIEM Deployment on Raspberry Pi (Debian ARM64)

This document walks through the full "from scratch" procedure for installing, configuring, and administering a complete native Wazuh stack (Indexer, Manager, Dashboard, Filebeat) directly on a Raspberry Pi (ARM64 architecture, Debian). It covers the step-by-step install, the systemd service sequencing, opening the firewall, hardening OpenSearch credentials, and fully resolving the Filebeat log-shipping pipeline.




## 1. Adding Repositories & Installing Binaries

The first step is preparing the Debian environment to pull the official Wazuh packages built for ARM64.

### 1.1 Add the GPG Key and Official Repository

```
curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | gpg --no-default-keyring \
  --keyring gnupg-ring:/usr/share/keyrings/wazuh.gpg --import && chmod 644 /usr/share/keyrings/wazuh.gpg
echo "deb [signed-by=/usr/share/keyrings/wazuh.gpg] https://packages.wazuh.com/4.x/apt/ stable main" | \
  tee -a /etc/apt/sources.list.d/wazuh.list
apt-get update
```

### 1.2 Install the Stack Components

Install the foundational building blocks in sequence: the indexer (OpenSearch), the analysis engine (Manager), the web interface (Dashboard), and the log collector/shipper (Filebeat).

```
apt-get install wazuh-indexer
apt-get install wazuh-manager
apt-get install wazuh-dashboard
apt-get install filebeat
```

## 2. Configuring the Log Shipper (Filebeat)

To ship alerts generated locally by the manager to the OpenSearch indexer, Filebeat is required. Without careful configuration, no logs will appear in the interface.

### 2.1 Download the Wazuh Filebeat Module and Configuration

Fetch Wazuh's default configuration and deploy the specific module responsible for parsing `alerts.json`.

```
curl -so /etc/filebeat/filebeat.yml \
  https://raw.githubusercontent.com/wazuh/wazuh/v4.14.6/extensions/filebeat/7.x/filebeat.yml
chmod 644 /etc/filebeat/filebeat.yml
curl -s https://packages.wazuh.com/4.x/filebeat/wazuh-filebeat-0.1.tar.gz | \
  tar -xvz -C /usr/share/filebeat/module
```

### 2.2 Deploy the JSON Template & Fix a Common Error

**Critical alert:** Filebeat absolutely requires a local structure template (`wazuh-template.json`) referenced in its configuration. If this file is missing, Filebeat will refuse to establish the connection (`onConnect callback failed`). Download the template to the right location:

```
curl -sL -o /etc/filebeat/wazuh-template.json \
  https://raw.githubusercontent.com/wazuh/wazuh/v4.14.6/extensions/elasticsearch/7.x/wazuh-template.json
```

### 2.3 Configure the Secure Connection

Edit Filebeat's configuration file:

```
nano /etc/filebeat/filebeat.yml
```

Update the `output.elasticsearch` section to point at the local indexer over HTTPS with your admin credentials (skipping strict certificate validation in this lab environment):

```
output.elasticsearch:
  hosts: ["https://127.0.0.1:9200"]
  protocol: "https"
  username: "admin"
  password: "YOUR_PASSWORD"
  ssl.verification_mode: "none"
```

## 3. Enabling and Starting Services (systemd)

Once every binary and configuration is in place, tell systemd to enable all daemons at boot and start the associated services.

```
systemctl daemon-reload

# Wazuh Indexer
systemctl enable wazuh-indexer
systemctl start wazuh-indexer

# Filebeat
systemctl enable filebeat
systemctl start filebeat

# Wazuh Manager
systemctl enable wazuh-manager
systemctl start wazuh-manager

# Wazuh Dashboard
systemctl enable wazuh-dashboard
systemctl start wazuh-dashboard
```

## 4. Firewall Configuration & Security (UFW)

To allow communication between agents and the manager, as well as access to the web interface (on port 8443), UFW must be adjusted.

```
ufw allow 8443/tcp comment "Wazuh Dashboard"
ufw allow 1514/tcp comment "Agent to Manager logs"
ufw allow 1515/tcp comment "Agent enrollment"
ufw reload
```

## 5. Configuring OpenSearch Credentials

The default `admin` user requires a secure password to be set within the indexer.

**1. Generate the hash** (example, for `YOUR_PASSWORD`):

```
/usr/share/wazuh-indexer/plugins/opensearch-security/tools/hash.sh -p "YOUR_PASSWORD"
```

**2. Edit the internal users file:**

```
nano /etc/wazuh-indexer/opensearch-security/internal_users.yml
```

Replace the `hash` value under the `admin` user with the generated string.

**3. Apply the configuration (security init):**

```
/usr/share/wazuh-indexer/bin/indexer-security-init.sh
```

## 6. Manually Injecting the Index Template

On first login, the Dashboard may display the error `No template found for index-pattern wazuh-alerts-*`. The template must be injected via the legacy API to work around OpenSearch's recent restrictions (which reject the `order` field on the modern API).

```
# Download the official template (e.g. v4.14.6)
curl -sL -o /tmp/wazuh-template.json \
  https://raw.githubusercontent.com/wazuh/wazuh/v4.14.6/extensions/elasticsearch/7.x/wazuh-template.json

# Inject via the legacy API
curl -k -u admin:YOUR_PASSWORD -X PUT "https://127.0.0.1:9200/_template/wazuh" \
  -H "Content-Type: application/json" -d @/tmp/wazuh-template.json

# Create the initial index
curl -k -u admin:YOUR_PASSWORD -X PUT "https://127.0.0.1:9200/wazuh-alerts-4.x-$(date +%Y.%m.%d)"
```

## 7. Architecture Trap: Manager / Agent Conflict

**Never** run `apt install wazuh-agent` on the server hosting the manager. The packages conflict over the `/var/ossec` directory, which will uninstall the manager.

The `wazuh-manager` service natively embeds agent `000` for its own self-monitoring. If you make this mistake, simply reinstall the manager — configuration files (`.conf`) are preserved:

```
apt install -y wazuh-manager
systemctl daemon-reload && systemctl start wazuh-manager
```

## 8. Validating the Detection Chain & Live Threat Hunting

To validate the full detection chain end to end:

### 8.1 Test the Filebeat Shipping Path

First confirm the connection between the collector and the database is fully established:

```
filebeat test output
```

If the test shows `talk to server... OK`, Filebeat is shipping alerts correctly. You can also query the local indexer directly to watch the index fill up:

```
curl -k -u admin:YOUR_PASSWORD -X GET "https://127.0.0.1:9200/_cat/indices?v"
```

The `docs.count` value for your `wazuh-alerts-*` index should be greater than 0.

### 8.2 Simulate an Event

1. Log in to `https://<PI_IP>:8443`.
2. Go to **Threat Hunting** (or **Discover** on the `wazuh-alerts-*` index).
3. Generate an invalid authentication alert from a remote machine:

```
ssh nonexistent_user@<PI_IP>
```

The alert will show up instantly in the event stream, confirming your local SOC is operational.